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HOST COMPANY'S VERIFICATION STATEMENT

VERIFICATON STATEMENT

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1.0 Abstract

This project focuses on the development of two interactive dashboards using Microsoft Power BI, namely the Environmental, Social, and Governance (ESG) Dashboard and the Enterprise Mobile Customer Experience (EMCE) Dashboard. The primary objective of this project is to enhance data visibility, improve monitoring capabilities, and support data-driven decision-making within the organization through effective data visualization techniques.

In modern organizations, large volumes of operational and sustainability-related data are generated continuously. However, the challenge lies in transforming this raw data into meaningful insights that can support strategic and operational decisions. Traditional reporting methods are often static, time-consuming, and lack interactivity, making it difficult for stakeholders to gain real-time insights. Therefore, this project addresses the need for a dynamic and interactive dashboard solution that enables efficient data analysis and visualization.

The ESG dashboard is designed to provide a centralized platform for monitoring sustainability-related indicators, including environmental performance, governance practices, and social responsibility metrics. This supports organizations in improving transparency, enhancing accountability, and aligning with sustainability reporting requirements. On the other hand, the EMCE dashboard focuses on monitoring mobile network performance and customer experience indicators such as call setup success rate, drop call rate, streaming latency, and streaming throughput. These metrics are essential for evaluating service quality and ensuring customer satisfaction in the telecommunications sector.

Due to confidentiality constraints, simulated datasets were generated using Microsoft Excel to replicate realistic operational scenarios. The data was then integrated into Microsoft Fabric and processed using Power Query for cleaning and transformation. Analytical measures were created using Data Analysis Expressions (DAX) to support advanced data analysis within the dashboards.

The final outcome demonstrates how business intelligence tools can effectively transform structured datasets into meaningful visual insights. The dashboards provide interactive features such as filtering, drill-down analysis, and geographical visualization, enabling users to identify trends, detect issues, and make informed decisions efficiently.

2.0 Introduction

2.1 Background of study

In today's highly digitalized business environment, organizations generate and process vast amounts of operational data on a daily basis. The rapid advancement of digital technologies has significantly increased the volume, velocity, and variety of data available to organizations across multiple industries. As a result, companies are increasingly relying on data-driven decision-making processes to maintain competitiveness and improve operational efficiency. However, the ability to extract meaningful insights from large datasets requires advanced analytical tools capable of transforming raw data into useful information.

Business intelligence (BI) systems have emerged as a critical component in modern data management strategies. BI tools allow organizations to collect, process, analyze, and visualize data in a structured manner, enabling stakeholders to make informed decisions based on real-time insights. Data visualization platforms such as Microsoft Power BI provide organizations with interactive dashboards and reporting capabilities that simplify complex data structures into easily understandable visual formats (*Microsoft, 2023*). Through graphical elements such as charts, tables, maps, and performance indicators, users can quickly identify trends, patterns, and anomalies within datasets that would otherwise be difficult to detect through traditional data analysis methods.

One of the major advantages of business intelligence dashboards is their ability to centralize data from multiple sources into a single visualization platform. Instead of manually collecting and compiling data from different systems, organizations can integrate various datasets into a unified analytical environment. This centralized approach improves data accessibility, enhances transparency, and allows decision-makers to monitor performance indicators more efficiently. Furthermore, interactive dashboard features such as filters, slicers, and drill-down functions enable users to explore data dynamically and obtain deeper analytical insights.

Within the telecommunications industry, data analytics plays a particularly important role in monitoring operational performance and ensuring service reliability. Telecommunications networks generate massive volumes of operational data related to network usage, infrastructure performance, and customer interactions. Monitoring these metrics is essential for maintaining service quality and ensuring customer satisfaction. Network performance indicators such as call setup success rates, drop call rates, network latency, and data throughput are commonly used to

evaluate the performance of mobile network services (*Johansson & Kidron, 2017*). By continuously monitoring these indicators, telecommunications providers can identify service disruptions, detect network inefficiencies, and implement corrective measures to improve service delivery.

Customer experience has also become a key competitive factor within the telecommunications sector. As mobile connectivity becomes increasingly integrated into everyday life, customers expect reliable network performance and seamless digital experiences. Any disruption in service quality, such as poor network coverage or high latency, can negatively impact customer satisfaction and lead to increased customer churn. Consequently, telecommunications companies must implement advanced monitoring systems that allow them to track customer experience indicators and respond quickly to emerging issues. Analytical dashboards that provide real-time visibility into network performance can help organizations proactively manage service quality and maintain customer trust.

In addition to operational monitoring, modern organizations are also facing increasing pressure to demonstrate responsible and sustainable business practices. Environmental, Social, and Governance (ESG) considerations have become an important aspect of corporate strategy and performance evaluation. ESG frameworks provide a structured approach for organizations to measure and report their environmental impact, social contributions, and governance practices. Investors, regulators, and stakeholders are increasingly using ESG metrics to assess the sustainability performance and ethical standards of companies.

ESG reporting requires organizations to collect and analyze data related to various sustainability indicators, including energy consumption, carbon emissions, workforce diversity, and governance practices. Traditionally, ESG reporting has been conducted through static reports and manual data collection processes. However, as ESG reporting requirements become more complex, organizations are increasingly adopting digital monitoring systems to improve the accuracy and efficiency of sustainability data management. Digital dashboards enable organizations to track ESG metrics in real time and provide stakeholders with transparent access to sustainability performance data (*Cui, 2025*).

Research has also highlighted the role of digital transformation in enhancing ESG management within the telecommunications sector. Telecommunications infrastructure consumes significant amounts of energy and resources, making sustainability monitoring an important component of industry operations. Digital monitoring systems allow telecommunications companies to evaluate their environmental impact, track sustainability performance indicators, and align their

operations with global sustainability standards (*Vetrova et al., 2023*). By integrating ESG monitoring with operational data analytics, organizations can gain a more comprehensive understanding of both their business performance and their sustainability responsibilities.

The integration of business intelligence tools with ESG monitoring and customer experience analytics provides organizations with a comprehensive data management solution. By combining operational performance indicators with sustainability metrics within a centralized dashboard environment, organizations can improve data visibility, enhance decision-making processes, and strengthen corporate transparency. Such systems not only support internal operational monitoring but also contribute to external reporting and stakeholder engagement.

Therefore, the development of interactive dashboards that integrate ESG monitoring and customer experience analytics represents an important step toward improving organizational data management practices. These dashboards allow organizations to convert raw datasets into meaningful insights that support strategic decision-making, operational efficiency, and long-term sustainability objective

2.2 Problem Statement

Despite the increasing availability of digital data, many organizations continue to face challenges in effectively managing and analyzing large volumes of information. Operational data, customer experience metrics, and sustainability indicators are often stored across multiple systems and databases within an organization. This fragmented data environment can make it difficult for stakeholders to obtain a comprehensive view of organizational performance. Without a centralized data management platform, decision-makers may need to rely on manual data compilation processes or multiple disconnected reports to obtain the information they require.

Manual data processing methods are often time-consuming and prone to human error. When organizations rely on static reports or spreadsheets for performance monitoring, the data may quickly become outdated or inconsistent. This lack of real-time visibility can hinder an organization's ability to respond quickly to operational issues or emerging trends. As a result, organizations may experience delays in identifying performance problems, which can negatively affect operational efficiency and strategic planning.

In the telecommunications industry, the ability to monitor network performance and customer experience in real time is particularly critical. Telecommunications services depend heavily on reliable network infrastructure and consistent service quality. If network issues such as high latency, low throughput, or dropped calls occur, they can significantly impact the user experience. Without effective monitoring systems, these issues may not be detected until they have already affected a large number of customers.

Customer experience is one of the key determinants of customer loyalty and retention in the telecommunications sector. Research indicates that service quality and customer satisfaction are closely linked to network performance and service reliability (*Mogire et al., 2024*). When customers encounter repeated service disruptions or poor network performance, they may become dissatisfied and choose to switch to alternative service providers. Therefore, telecommunications companies must implement advanced monitoring systems that allow them to track service performance indicators and detect potential problems before they escalate.

Another challenge faced by organizations relates to sustainability reporting and ESG performance monitoring. ESG reporting often requires data collection from multiple departments within an organization, including environmental management teams, human resources departments, and corporate governance units. The process of gathering and consolidating sustainability data can be complex and time-consuming, particularly when the

data is stored in different formats and systems.

Without automated monitoring tools, ESG reporting processes may rely heavily on manual data collection and reporting methods. This can lead to inconsistencies in reporting and difficulties in maintaining accurate records of sustainability performance indicators. In addition, the lack of centralized monitoring systems may limit an organization's ability to track ESG performance trends over time. Effective sustainability management requires organizations to continuously evaluate their environmental and social impact and make adjustments to their operational strategies.

Research has shown that transparent sustainability reporting can improve organizational credibility and strengthen stakeholder trust (*Buallay & Al Marri, 2022*). Investors and regulatory bodies increasingly expect organizations to provide accurate and transparent ESG disclosures. However, without efficient data monitoring systems, organizations may struggle to provide timely and reliable sustainability reports.

Given these challenges, there is a clear need for integrated digital solutions that can support both operational monitoring and sustainability data management. Centralized dashboards provide an effective solution by integrating multiple data sources into a unified analytical platform. Through interactive data visualizations, stakeholders can obtain a comprehensive overview of performance indicators and identify areas that require improvement.

By implementing centralized dashboard systems, organizations can improve data accessibility, reduce reliance on manual reporting processes, and enhance their ability to make data-driven decisions. In the telecommunications industry, such systems can support network performance monitoring, customer experience analysis, and sustainability reporting within a single analytical environment.

Therefore, the development of interactive dashboards for monitoring ESG performance and enterprise mobile customer experience can help address these challenges by providing organizations with a centralized platform for data visualization, operational monitoring, and strategic decision-making.

3.0 OBJECTIVE

The main objective of this Student Industrial Project is to design and develop business intelligence dashboards that support operational monitoring and sustainability reporting.

The specific objectives of this project include:

1. To develop an ESG dashboard that monitors sustainability performance indicators within the organization.
2. To develop an Enterprise Mobile Customer Experience (EMCE) dashboard that visualizes mobile network performance and service quality metrics.
3. To transform and integrate datasets using Microsoft Excel, Microsoft Fabric, and Power Query.
4. To build a semantic data model within Microsoft Power BI for accurate data analysis.
5. To design interactive visualizations that allow users to monitor key performance indicators efficiently.

3.1 Scope of work

The scope of this project focuses on the design, development, and implementation of two interactive dashboards using Microsoft Power BI, the Environmental, Social, and Governance (ESG) Dashboard and the Enterprise Mobile Customer Experience (EMCE) Dashboard. These dashboards are developed to support data visualization, performance monitoring, and analytical decision-making by transforming structured datasets into meaningful graphical representations.

The ESG dashboard is designed to monitor and visualize sustainability-related performance indicators within the organization. ESG metrics are increasingly important for organizations as they provide insights into environmental impact, social responsibility initiatives, and corporate governance practices. The dashboard aims to centralize ESG related information into a single visualization platform, allowing stakeholders to track sustainability indicators more efficiently and evaluate organizational performance in relation to ESG objectives. By consolidating ESG data into an interactive dashboard, the system supports improved transparency, reporting efficiency, and sustainability monitoring.

In addition to ESG monitoring, the project also involves the development of the Enterprise Mobile Customer Experience (EMCE) dashboard. This dashboard focuses on monitoring key operational indicators related to mobile network performance and customer experience within the telecommunications environment. The EMCE dashboard visualizes network performance metrics such as call setup success rates, drop call rates, streaming latency, and streaming throughput. These indicators provide valuable insights into the overall quality of network services and allow stakeholders to identify areas where network performance may require improvement. Through the visualization of these metrics, the dashboard helps support operational monitoring and enables management teams to respond more effectively to potential service disruptions.

Due to the confidential nature of real corporate data, the project utilizes simulated datasets created using Microsoft Excel. These datasets are designed to mimic realistic operational scenarios while ensuring that no sensitive organizational information is disclosed. The simulated data includes structured values that represent ESG performance indicators and mobile network performance metrics. The use of dummy datasets ensures compliance with organizational data governance policies while still allowing the development and demonstration of functional analytical dashboards.

After the datasets are created, they are uploaded into Microsoft Fabric, which serves as the data integration and storage environment for the project. Microsoft Fabric allows the datasets to be managed and structured before they are connected to the Power BI environment for analysis and visualization. This step ensures that the data is stored in a centralized platform that supports efficient data retrieval and processing.

Once the datasets are integrated into Microsoft Fabric, Power Query is used to perform data cleaning and transformation processes. Power Query allows the datasets to be prepared for analysis by removing inconsistencies, restructuring columns, and ensuring that the data format is compatible with the analytical model used in Power BI. This stage is important for improving data quality and ensuring that the visual outputs generated by the dashboards accurately represent the underlying dataset.

Following the data transformation stage, a semantic data model is developed within Power BI. This process involves creating relationships between different data tables to ensure that the datasets can be analyzed effectively. Data modeling also enables the creation of calculated metrics using Data Analysis Expressions (DAX), which allows the dashboard to generate aggregated performance indicators and key analytical measures. These calculations are necessary to support the visualization of performance metrics and provide meaningful insights to dashboard users.

The dashboards developed within the scope of this project incorporate a variety of data visualization components. These include Key Performance Indicator (KPI) cards, bar charts, line charts, pie charts, tables, matrix tables, and geographical maps. Each visualization element is designed to represent specific performance indicators in a clear and intuitive manner. KPI cards provide high-level summaries of important metrics, while charts and graphs illustrate trends and comparisons across different datasets.

Interactive features are also incorporated into the dashboards to enhance user experience and analytical capabilities. Filters, slicers, and interactive controls allow users to customize the data view based on specific parameters such as time periods, geographic regions, or performance categories. These interactive features enable users to explore the data more dynamically and obtain deeper insights into operational trends and performance patterns.

Another important aspect within the scope of this project is the design and layout of the dashboards. The dashboard interface is structured to ensure clarity, usability, and effective information presentation. Visual elements are organized in a logical layout so that users can easily navigate through the dashboard and identify important information without confusion. Proper dashboard design helps improve readability and ensures that key performance indicators

are clearly highlighted.

In addition to dashboard development, the project scope also includes testing and validation of the dashboards. System testing is conducted to ensure that the dashboards function correctly and that all visual components display accurate data. This process involves verifying data relationships, validating DAX calculations, and ensuring that filters and interactive elements work as intended. The testing phase ensures that the dashboards provide reliable analytical outputs and function efficiently within the Power BI environment.

Overall, the scope of this project covers multiple stages of business intelligence development, including data preparation, data integration, data modeling, dashboard design, visualization development, and system validation. Through these stages, the project demonstrates how business intelligence tools can be used to transform raw datasets into centralized analytical platforms that support operational monitoring and data-driven decision-making.

The final outcome of the project is the successful implementation of two functional dashboards that provide stakeholders with a centralized view of ESG performance metrics and mobile customer experience indicators. These dashboards serve as analytical tools that support performance monitoring, operational analysis, and strategic decision-making within the organization.

4.0 LITERATURE REVIEW

4.1 Business Intelligence and Data Visualisation

Business Intelligence (BI) systems have become an essential component of modern organizational data management strategies. As organizations generate increasingly large volumes of data through digital operations, the ability to transform raw datasets into meaningful insights has become critical for effective decision-making and strategic planning. BI systems allow organizations to collect, integrate, analyze, and visualize data in ways that support operational monitoring and analytical evaluation. By using data analytics and visualization techniques, BI tools enable organizations to identify patterns, monitor key performance indicators (KPIs), and detect emerging trends within complex datasets. This capability allows organizations to move beyond traditional data reporting methods and adopt more proactive and data-driven management practices (*Microsoft, 2023*).

One of the core functions of business intelligence platforms is the transformation of raw data into structured information that can be easily interpreted by decision-makers. Through the use of data visualization techniques, BI dashboards present analytical outputs in graphical formats such as charts, graphs, tables, and maps. These visual representations simplify complex datasets and enable users to quickly understand relationships between different variables. According to *Microsoft (2023)*, visualization tools integrated within BI platforms help users analyze data more efficiently by converting numerical values and statistical outputs into visual elements that support intuitive interpretation.

Modern data visualization platforms have significantly evolved with the advancement of digital technologies. Tools such as Microsoft Power BI provide interactive interfaces that allow users to explore datasets dynamically and customize data views based on specific analytical needs. Interactive dashboard features, including filters, slicers, drill-down functions, and cross-filtering capabilities, enable users to perform deeper analysis without requiring advanced technical expertise. These functionalities allow users to interact with datasets in real time, facilitating a more comprehensive understanding of organizational performance indicators (*Microsoft, 2023*).

In addition to supporting data analysis, business intelligence dashboards also play a crucial role in improving organizational transparency and communication. Centralized dashboards provide stakeholders with a unified view of operational data by integrating information from multiple sources into a single platform. This centralized approach eliminates data silos and ensures that different departments within an organization are working with consistent and up-to-date information. By establishing a centralized data environment, organizations can create a "single

source of truth" that enhances data reliability and reduces inconsistencies in reporting (*Buallay & Al Marri, 2022*).

The ability to centralize data is particularly important for organizations that operate in complex and data-intensive industries such as telecommunications. Telecommunications companies manage vast amounts of operational data related to network performance, service quality, customer interactions, and infrastructure utilization. Effective management of this data requires advanced analytical tools that can aggregate information from different systems and present it through a unified visualization interface. BI dashboards allow telecommunications providers to monitor operational performance metrics, evaluate service delivery, and identify potential performance issues before they impact customers (*Johansson & Kidron, 2017*).

Furthermore, centralized dashboards improve collaboration among stakeholders by making critical information easily accessible to various departments within an organization. When data is integrated into a centralized BI platform, managers, analysts, and operational teams can access the same information simultaneously, enabling more coordinated decision-making processes. This shared data environment enhances communication across departments and allows organizations to respond more quickly to operational challenges or emerging opportunities (*Microsoft, 2023*).

The role of BI dashboards has also expanded to support sustainability monitoring and corporate governance initiatives. Organizations increasingly rely on digital dashboards to track Environmental, Social, and Governance (ESG) indicators and evaluate their sustainability performance. Research suggests that digital monitoring platforms allow organizations to integrate sustainability metrics with operational data, enabling a more comprehensive evaluation of corporate performance (*Cui, 2025*). By incorporating ESG indicators into centralized dashboards, organizations can monitor sustainability goals, track environmental impact, and ensure compliance with regulatory reporting standards.

Digital transformation within the telecommunications sector has further emphasized the importance of advanced data analytics and visualization tools. As telecommunications networks become more complex and data-driven, organizations require sophisticated monitoring systems that provide real-time insights into network operations and service performance. Digital dashboards enable telecommunications companies to analyze operational metrics such as network latency, service reliability, and customer engagement levels. These insights support improved operational management and contribute to enhanced service quality (*Vetrova et al., 2023*).

In addition, sustainability considerations have become increasingly relevant in the telecommunications industry due to the environmental impact associated with large-scale network infrastructure and energy consumption. Monitoring sustainability indicators through digital dashboards allows organizations to evaluate the environmental efficiency of telecommunications networks and implement strategies for reducing operational impact. Research in telecommunications sustainability emphasizes the importance of monitoring network performance and environmental indicators simultaneously in order to support long-term sustainable development (*Pintor et al., 2025*).

Overall, business intelligence dashboards serve as powerful analytical tools that allow organizations to transform raw data into actionable insights through visualization and interactive analysis. By integrating multiple datasets into centralized platforms, BI systems enhance data accessibility, improve operational transparency, and support faster decision-making processes. The growing adoption of BI tools in various industries highlights the importance of digital analytics platforms in supporting modern data-driven organizational environments.

4.2 ESG Data Monitoring and Sustainability Reporting

Environmental, Social, and Governance (ESG) performance monitoring has become an increasingly important component of modern corporate sustainability strategies. In recent years, organizations across multiple industries have faced growing pressure from investors, regulators, and stakeholders to demonstrate responsible and transparent business practices. ESG frameworks provide a structured method for organizations to evaluate their environmental impact, social responsibility initiatives, and governance practices. As a result, organizations are increasingly expected to disclose sustainability-related metrics and integrate ESG considerations into their corporate strategies. Effective ESG monitoring enables companies to demonstrate accountability, improve transparency, and strengthen stakeholder trust by providing measurable evidence of their sustainability performance (*Buallay & Al Marri, 2022*).

Sustainability reporting has evolved significantly over the past decade as organizations move beyond traditional financial performance indicators to incorporate environmental and social metrics into their evaluation processes. ESG disclosure allows stakeholders to better understand how organizations manage environmental risks, promote ethical governance, and contribute to social development. Research has shown that organizations with transparent sustainability reporting practices are more likely to attract investors and maintain long-term business stability. In particular, companies within technology-driven sectors such as telecommunications must carefully monitor sustainability indicators due to their significant operational and environmental impact (*Buallay & Al Marri, 2022*).

The increasing complexity of ESG reporting has led organizations to adopt digital tools that can support efficient data collection, monitoring, and analysis. Digital platforms such as ESG dashboards provide organizations with centralized systems for tracking sustainability indicators in real time. These dashboards integrate data from multiple departments and operational systems, allowing organizations to monitor environmental impact, social responsibility metrics, and governance compliance indicators through a unified analytical interface. By automating data collection and visualization processes, ESG dashboards improve the efficiency and accuracy of sustainability reporting while reducing reliance on manual reporting methods (*Cui, 2025*).

The use of digital monitoring systems also allows organizations to move beyond static sustainability reports toward more dynamic and proactive sustainability management practices. Instead of evaluating ESG performance only during annual reporting periods, organizations can continuously monitor sustainability indicators and identify trends over time. This real-time

monitoring capability allows management teams to detect potential sustainability risks early and implement corrective actions before they escalate into larger operational or reputational issues. Consequently, digital ESG monitoring platforms contribute to improved corporate governance and strategic planning processes (*Cui, 2025*).

Digital transformation has further accelerated the adoption of ESG monitoring systems across various industries, including the telecommunications sector. Telecommunications companies operate extensive network infrastructures that require significant energy consumption and technological resources. As a result, the environmental impact of telecommunications operations has become an important consideration in corporate sustainability strategies. Research on digital transformation within the telecommunications industry highlights that organizations increasingly rely on advanced data analytics and monitoring systems to evaluate sustainability performance and optimize resource utilization (*Vetrova et al., 2023*).

The integration of digital monitoring tools with sustainability management practices enables telecommunications companies to analyze ESG metrics more effectively and incorporate sustainability considerations into operational decision-making processes. Through centralized monitoring systems, organizations can evaluate energy consumption, infrastructure sustainability, and operational efficiency across their network environments. These analytical insights allow companies to identify areas where operational improvements can reduce environmental impact while maintaining high levels of service performance (*Vetrova et al., 2023*).

In addition to environmental indicators, ESG monitoring within telecommunications networks also involves evaluating governance structures and social responsibility initiatives. Social metrics may include factors such as employee diversity, workplace safety, digital inclusion initiatives, and community engagement programs. Governance indicators focus on corporate transparency, ethical management practices, and regulatory compliance. Monitoring these indicators helps organizations ensure that their business operations align with international sustainability standards and ethical business practices (*Buallay & Al Marri, 2022*).

Another important area of ESG monitoring within the telecommunications industry relates to network sustainability and energy efficiency. Telecommunications networks require large-scale infrastructure, including data centers, base stations, and communication equipment, all of which consume significant amounts of energy. Monitoring sustainability indicators related to energy consumption, network efficiency, and infrastructure utilization allows organizations to evaluate the environmental impact of their operations. According to recent research, sustainability monitoring systems play a key role in enabling telecommunications companies to improve

energy efficiency and reduce operational carbon emissions (*Pintor et al., 2025*).

Furthermore, sustainability monitoring systems allow organizations to evaluate the long-term sustainability of their network infrastructure and technological investments. By analyzing operational data alongside environmental metrics, telecommunications providers can implement strategies that optimize resource usage while minimizing environmental impact. This integrated approach to sustainability management allows organizations to align operational performance with broader sustainability objectives (*Pintor et al., 2025*).

Overall, ESG monitoring has become a critical component of modern corporate management practices. Digital dashboards and analytics platforms enable organizations to integrate sustainability metrics with operational data, providing a comprehensive view of organizational performance. By adopting digital monitoring systems, organizations can improve transparency, enhance sustainability reporting processes, and support data-driven decision-making that aligns with long-term environmental and social responsibility goals (*Cui, 2025 & Vetrova et al., 2023*).

The growing importance of ESG monitoring across industries highlights the need for effective digital solutions that can support sustainability management and performance evaluation. Interactive dashboards that visualize ESG metrics allow organizations to track sustainability indicators more efficiently while ensuring that sustainability considerations remain integrated into strategic decision-making processes.

4.3 Customer Experience Analytics in Telecommunications

Customer experience analytics has become an increasingly important area of study within the telecommunications industry, particularly as digital connectivity becomes an essential component of modern daily life. Telecommunications providers deliver services that support voice communication, internet connectivity, video streaming, and mobile applications. Because these services are heavily dependent on network infrastructure and system performance, the quality of the customer experience is closely tied to the reliability and efficiency of telecommunications networks. As a result, organizations must continuously monitor service performance and customer interaction metrics to ensure that service quality meets user expectations and maintains competitive advantage in the telecommunications market (Johansson & Kidron, 2017).

Customer experience within telecommunications services is influenced by several operational and technical factors, including network reliability, data transmission speed, service availability, and application responsiveness. When customers encounter issues such as dropped calls, slow data connections, or service interruptions, their overall satisfaction with the service provider may decrease significantly. In competitive telecommunications markets, poor service performance can lead to customer dissatisfaction and increased customer churn. Therefore, telecommunications companies must implement effective monitoring systems that allow them to track service performance indicators and identify potential service disruptions before they negatively impact customers (Mogire et al., 2024).

Monitoring customer experience metrics allows telecommunications providers to evaluate how users interact with network services and digital platforms. Analytical indicators such as call setup success rates, drop call rates, network latency, and application response times provide valuable insights into the quality of service experienced by users. These performance metrics allow organizations to detect patterns in network performance and identify operational issues that may require technical intervention. For example, if network latency increases in a particular geographic area, telecommunications engineers can investigate potential infrastructure issues and implement corrective actions to improve service performance (Johansson & Kidron, 2017).

In addition to network performance indicators, customer experience analytics also involves analyzing user behavior and interaction patterns across digital platforms. Modern telecommunications services often include mobile applications, online customer service platforms, and digital support systems that allow users to manage their accounts and access services. By analyzing usage patterns and customer feedback data, organizations can better understand how customers interact with their services and identify areas where improvements

may be required. This information enables companies to optimize digital platforms and enhance overall user experience (*Williamson, 2016*).

Research on customer experience management within the telecommunications industry highlights the importance of maintaining high levels of service reliability and responsiveness. Studies have shown that improvements in service performance, such as reduced network latency and increased connection reliability, contribute significantly to higher levels of customer satisfaction. When customers experience consistent service quality, they are more likely to maintain long-term relationships with service providers and demonstrate greater brand loyalty (*Mogire et al., 2024*).

Furthermore, the telecommunications industry is increasingly adopting digital analytics platforms to support customer experience monitoring and performance evaluation. Analytical dashboards provide telecommunications providers with real-time visibility into operational performance and customer interaction metrics. Through the use of visual analytics tools, organizations can convert complex network performance data into graphical representations that are easier to interpret and analyze. These dashboards allow operational teams and management personnel to quickly identify performance trends, detect anomalies, and make informed decisions based on real-time data insights (*Microsoft, 2023*).

The use of centralized analytical dashboards also improves collaboration among different departments within telecommunications organizations. Network engineers, customer service teams, and management personnel can access the same operational data through a unified dashboard interface. This shared data environment ensures that all stakeholders have access to consistent information regarding service performance and customer experience indicators. As a result, organizations can coordinate their responses to operational issues more effectively and implement service improvements in a timely manner (*Johansson & Kidron, 2017*).

Another important benefit of customer experience analytics is its ability to support proactive service management strategies. Instead of reacting to customer complaints after service issues occur, telecommunications providers can use predictive analytics and monitoring tools to identify potential problems in advance. By analyzing trends in network performance metrics, organizations can detect early warning signs of network congestion, infrastructure failures, or service degradation. Early detection allows organizations to implement preventative measures that minimize service disruptions and maintain a consistent level of service quality (*Williamson, 2016*).

In addition, digital analytics platforms enable organizations to integrate multiple sources of operational and customer-related data into a single monitoring system. For example, network performance metrics, customer service feedback, application usage statistics, and service complaint records can all be analyzed together within an integrated dashboard environment. This holistic view of customer experience allows organizations to better understand the relationship between network performance and customer satisfaction outcomes (*Mogire et al., 2024*).

Overall, customer experience analytics plays a crucial role in helping telecommunications providers maintain service reliability and improve customer satisfaction. By leveraging advanced data analytics and visualization technologies, organizations can transform complex operational data into meaningful insights that support service quality management. Analytical dashboards enable organizations to monitor performance indicators continuously, detect potential service issues early, and implement corrective measures that enhance customer experience and maintain long-term customer loyalty (*Microsoft, 2023 & Mogire et al., 2024*).

The growing importance of customer experience analytics within the telecommunications sector highlights the need for effective data visualization platforms that support real-time monitoring and decision-making. Interactive dashboards that integrate network performance metrics and customer interaction data provide organizations with valuable analytical tools for improving service quality, optimizing network operations, and strengthening customer relationships.

5.0 METHODOLOGY

5.1 Overview of Methodology

The methodology adopted in this project follows a structured data analytics and business intelligence development framework. The framework is designed to transform raw datasets into meaningful visual insights through a systematic process consisting of several key stages. These stages include data preparation, data transformation, data modeling, dashboard development, and system validation. Each stage plays a critical role in ensuring that the final dashboards provide accurate, reliable, and interactive analytical outputs that support operational monitoring and decision-making.

The first stage of the methodology involves **data preparation**, which focuses on the creation and organization of datasets that will be used within the dashboard environment. Due to the confidentiality and security policies associated with real organizational data, the project utilizes simulated datasets that were generated using Microsoft Excel. These datasets were designed to replicate realistic operational scenarios and include structured values representing Environmental, Social, and Governance (ESG) performance indicators as well as Enterprise Mobile Customer Experience (EMCE) metrics. The simulated datasets contain key performance indicators such as network performance metrics, service reliability indicators, and sustainability-related measurements. The use of dummy datasets ensures that the development process remains compliant with organizational data protection policies while still allowing realistic dashboard implementation.

Following the creation of the datasets, the next stage involves **data integration and storage**. The prepared datasets were uploaded into the Microsoft Fabric environment, which acts as the centralized data platform used to manage and organize the project's datasets. Microsoft Fabric provides a scalable data infrastructure that supports efficient data integration, storage, and processing before the datasets are connected to the Power BI environment. By utilizing Microsoft Fabric as an intermediate data platform, the project demonstrates a modern data architecture approach in which datasets are managed within a centralized data ecosystem before being consumed by analytical tools.

After the datasets were successfully integrated into Microsoft Fabric, the next stage of the methodology focuses on **data transformation and data cleaning**. This process was conducted using Power Query within the Power BI environment. Power Query provides a powerful data transformation engine that allows developers to clean, restructure, and prepare datasets before

they are used for analytical modeling. During this stage, several data transformation operations were performed, including column restructuring, data formatting adjustments, removal of redundant data fields, and validation of data consistency. These transformation processes ensure that the datasets are structured correctly and optimized for analytical processing within the dashboard environment.

Once the data transformation process was completed, the methodology proceeded to the **data modeling stage**. Data modeling is an essential component of business intelligence development as it defines the relationships between different datasets and enables meaningful analytical queries. In this project, a semantic data model was developed within Power BI by establishing relationships between multiple data tables. These relationships allow the system to connect related datasets and generate aggregated insights from different sources of information. The semantic model serves as the analytical backbone of the dashboard system, ensuring that all data relationships are logically structured and that analytical queries can be performed efficiently.

In addition to establishing relationships between tables, the data modeling stage also involved the creation of calculated metrics using **Data Analysis Expressions (DAX)**. DAX is a formula language used within Power BI to perform advanced calculations and generate key performance indicators. Through the use of DAX formulas, several analytical measures were created to support dashboard visualizations. These measures allow the system to calculate aggregated values, performance ratios, and trend indicators based on the underlying datasets. The implementation of DAX calculations enhances the analytical capabilities of the dashboard and allows users to gain deeper insights into the operational metrics being monitored.

Following the completion of data modeling and calculation development, the project moved into the **dashboard development stage**. During this phase, interactive dashboards were designed and developed using Microsoft Power BI visualization components. A variety of visual elements were incorporated into the dashboards, including Key Performance Indicator (KPI) cards, bar charts, line charts, pie charts, matrix tables, and geographical maps. Each visualization component was carefully selected to represent specific data metrics in a clear and intuitive manner. The dashboard layout was structured to ensure that important performance indicators are prominently displayed and easily accessible to users.

Interactive features were also integrated into the dashboards to enhance user experience and analytical flexibility. These features include slicers, filters, and interactive drill-down

capabilities that allow users to customize the data view based on selected parameters such as time periods, geographical locations, or specific performance categories. These interactive functions enable users to explore the datasets dynamically and obtain more detailed insights into performance trends and operational metrics.

The final stage of the methodology involves **system validation and testing**. This phase ensures that the developed dashboards function correctly and that all visual outputs accurately reflect the underlying datasets. Several validation processes were conducted during this stage, including verification of data relationships, validation of DAX calculations, and testing of interactive dashboard components such as filters and slicers. The testing process also ensured that all visualizations display accurate and consistent information across different data views.

Overall, the methodology adopted in this project demonstrates a complete business intelligence development workflow that integrates data preparation, transformation, modeling, visualization, and validation processes. By following this structured development approach, the project successfully transforms simulated datasets into interactive analytical dashboards that support performance monitoring, sustainability tracking, and customer experience analysis. The resulting dashboards provide stakeholders with a centralized visualization platform that enhances data accessibility, improves operational transparency, and supports data-driven decision-making.

5.2 Tools and technology used

The development of the ESG and Enterprise Mobile Customer Experience (EMCE) dashboards utilized a range of modern data analytics and business intelligence technologies to ensure an efficient and structured analytical workflow. These technologies were selected to support key processes including data preparation, transformation, modeling, and visualization within a centralized analytical environment. The project integrates Microsoft Excel for dataset generation, Microsoft Fabric for data integration and storage, Power Query for data transformation, and Microsoft Power BI for the development of interactive dashboards and visual analytics (*Microsoft, 2023*). The integration of these tools enables the effective processing and visualization of operational data, supporting improved data accessibility and analytical insight generation.

No	Tool / Resource	Description / Usage
1.	Microsoft Power BI	Used for dashboard development and data visualization.
2.	Microsoft Excel	Used to create dummy datasets for simulation purposes.
3.	Power Query	Used for data transformation and cleaning.
4.	DAX (Data Analysis Expression)	Used for creating calculated measures and analytical metrics.
5.	Microsoft Fabrics	Used for data storage and integration before connecting to Power BI.
6.	Python Notebook	Used for converting excel file into csv format
7.	Big Query Storage	Used for storing data in raw format and schedule refresh for every month new data.

Table 1: Tools and Resources Utilized

Tools Description:

1. Microsoft Power BI is used as the main platform for developing interactive dashboards and data visualizations in this project. The tool allows large datasets to be transformed into graphical insights through charts, KPI cards, tables, and maps. Power BI also enables users to interact with the dashboards using filters, slicers, and drill-down features to explore data more effectively (*Microsoft, 2023*).
2. Microsoft Excel is used to create simulated datasets for the ESG and Enterprise Mobile Customer Experience (EMCE) dashboards. The datasets include structured values representing key performance indicators such as sustainability metrics and network performance indicators. These datasets are used for development and testing purposes since real organizational data cannot be accessed due to confidentiality restrictions.
3. Power Query is used as the data transformation and data preparation tool within the Power BI environment. It allows datasets to be cleaned, formatted, and transformed before they are used for analytical modeling. Functions such as column restructuring, data filtering, and data type adjustments help ensure that the data is properly structured for dashboard development.
4. DAX is a formula language used in Power BI to create calculated measures and analytical metrics. It allows the system to perform advanced calculations such as aggregations, ratios, and time-based comparisons within the data model. By using DAX, the dashboards can generate meaningful performance indicators that support data-driven analysis.
5. Microsoft Fabric is used as the centralized data platform for storing and managing the project datasets before connecting them to Power BI. The platform allows datasets to be integrated, organized, and accessed within a unified data environment. Using Microsoft Fabric also demonstrates a modern data architecture approach where data processing and analytics are connected within a single ecosystem.
6. Python Notebook is used to convert Excel datasets into CSV format for compatibility with the data storage system. The notebook environment allows simple data processing tasks such as file conversion and dataset preparation to be performed efficiently. This step ensures that the datasets can be easily uploaded and processed within the data pipeline.
7. BigQuery Storage is used to store the datasets in their raw format before being processed for visualization. The storage system supports structured data management and enables scheduled data refresh processes to update the datasets periodically. In this project, the refresh schedule is configured monthly to simulate the integration of new data into the analytical system.

5.3 System Architecture

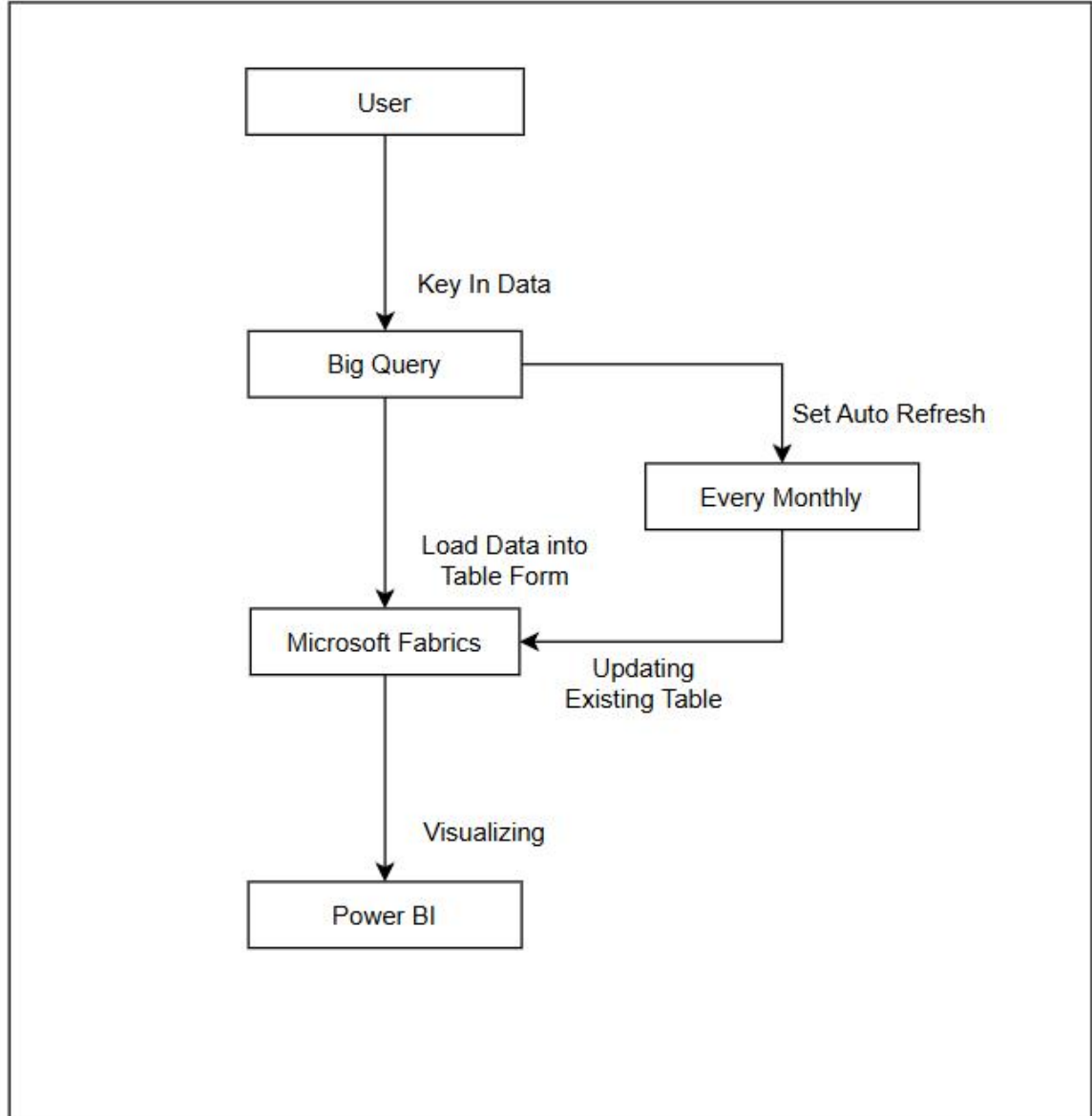


Figure 1: System architecture for the ESG Dashboard, illustrating data flow, setting auto refresh, and updating table.

Figure 1 illustrates the system architecture used for the Environmental, Social, and Governance (ESG) dashboard developed in this project. The architecture demonstrates how data flows from the initial data input stage until it is visualized in the Power BI dashboard. The process begins with the user, who manually inputs the simulated ESG datasets into the system. These datasets are stored in BigQuery, which acts as the primary storage environment for the raw data.

Once the data is stored in BigQuery, the system is configured to perform automated data refresh processes on a monthly basis. This scheduled refresh ensures that the stored datasets can be periodically updated and maintained without requiring continuous manual intervention. The refreshed data is then loaded into Microsoft Fabric, where it is converted into structured table formats that can be used for further data processing and analysis.

Microsoft Fabric functions as the centralized data integration and processing environment within the system architecture. In this stage, the data is prepared and organized so that it can be accessed efficiently by the visualization platform. After the data preparation process is completed, the processed datasets are connected to Microsoft Power BI, which is responsible for transforming the data into interactive visual dashboards.

Power BI serves as the final visualization layer of the system, where ESG performance indicators are presented through various visual components such as charts, KPI cards, and tables. This architecture enables centralized monitoring of sustainability metrics and supports data-driven decision-making by presenting complex datasets in an intuitive and accessible format.

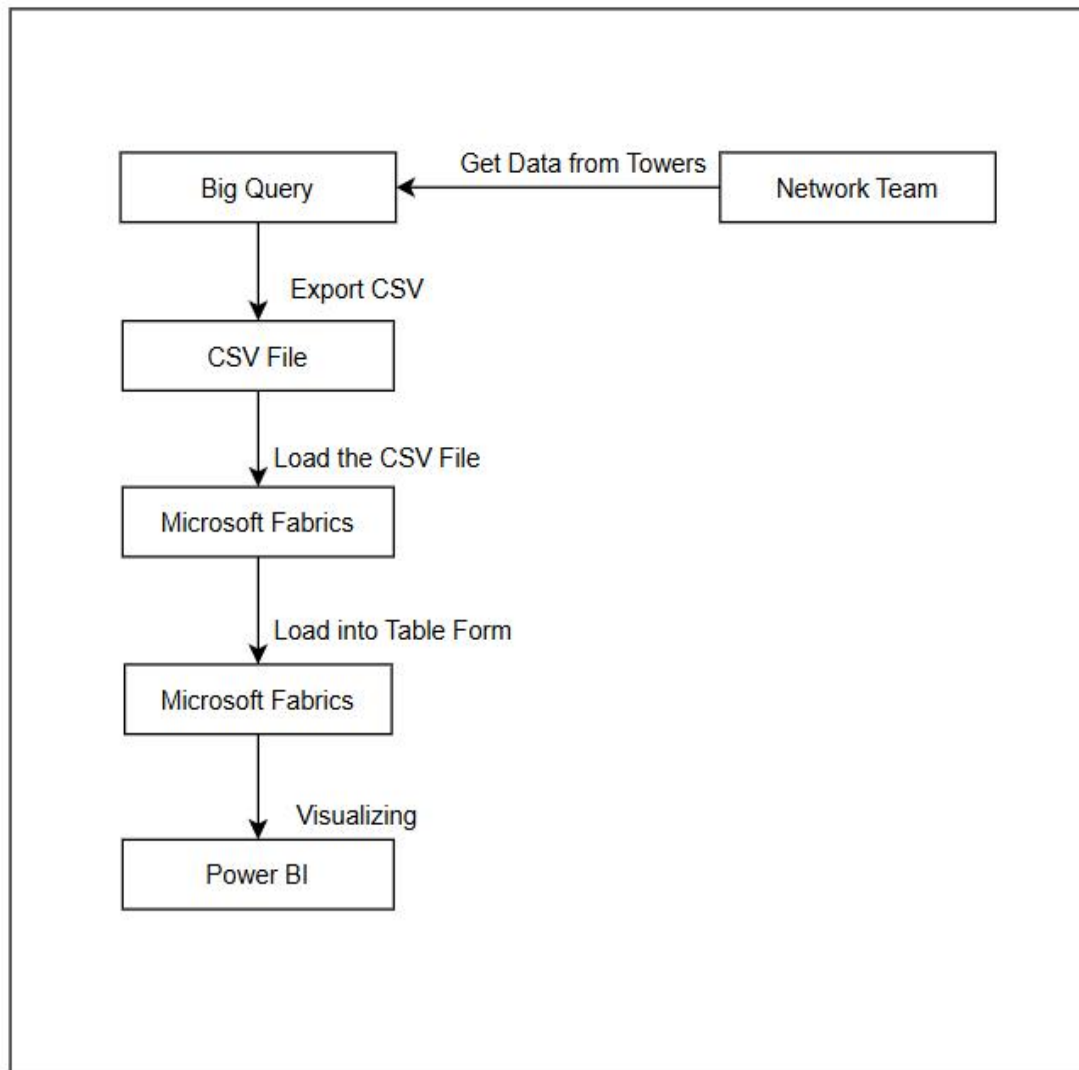


Figure 2: System architecture for the EMCE Dashboard, illustrating data flow, getting data from network towers, and uploading and converting data into table format

Figure 2 presents the system architecture implemented for the Enterprise Mobile Customer Experience (EMCE) dashboard. This architecture illustrates how mobile network performance data is collected, processed, and transformed into analytical insights for monitoring customer experience within the telecommunications network.

The process begins with the Network Team, which collects operational network performance data from telecommunications towers and network infrastructure. These data points include performance metrics related to network connectivity, service reliability, and customer usage patterns. The collected data is then stored and processed within BigQuery, which acts as the initial data repository for network-related datasets.

After the data is stored in BigQuery, it is exported into CSV file format, allowing the data to be transferred and integrated into the next stage of the data pipeline. The exported CSV files are then uploaded into Microsoft Fabric, which functions as the central data platform used for data integration and preparation.

Within Microsoft Fabric, the uploaded CSV files are transformed into structured table formats, allowing the datasets to be organized and prepared for analytical processing. This transformation step ensures that the data is structured correctly and can be efficiently connected to the visualization platform.

Once the data preparation process is completed, the datasets are connected to Microsoft Power BI, where the EMCE dashboard is developed. Power BI visualizes the network performance metrics through interactive charts, maps, and KPI indicators. These visualizations allow stakeholders to monitor important performance indicators such as network coverage quality, call success rates, and service reliability across different geographic areas.

Overall, the system architecture supports an integrated data pipeline that connects network data collection, data storage, data processing, and data visualization. By leveraging BigQuery, Microsoft Fabric, and Power BI, the system enables centralized monitoring of mobile customer experience metrics and provides stakeholders with actionable insights for improving network performance and service quality.

5.4 Data Acquisition and Storage Design

To support ESG analysis, two primary datasets were used, namely the SAP Latest table and the Non-SAP Data table. These datasets represent different sources of electricity consumption data and are integrated within the Power BI environment to enable comprehensive analysis.

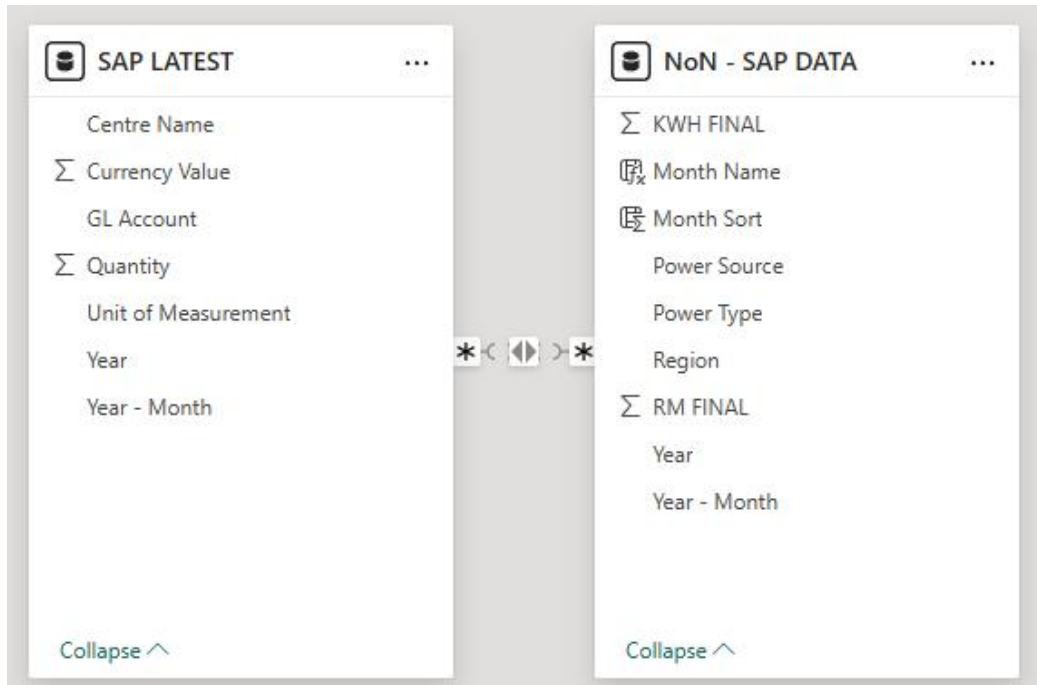


Figure 3 SAP and Non-SAP Semantic Models

Within the Non-SAP Data table, additional data transformation was performed to improve usability and ensure proper data organization. Two new fields were created, namely Month Name and Month Sort. The Month Name column is used to display the month in a readable format (e.g. January to December), while the Month Sort column ensures that the months are arranged in the correct chronological order rather than alphabetical order. This improves the accuracy and readability of time-based visualizations.

To establish a connection between the SAP and Non-SAP datasets, a relationship was created using the Year–Month field in both tables. This relationship is defined as a many-to-many relationship, as multiple records in each table may correspond to the same Year–Month value. This approach allows both datasets to be aligned based on a common time dimension, enabling combined analysis and consistent filtering across dashboards.

By integrating these two datasets, the model allows users to analyze electricity consumption data from multiple sources simultaneously, providing a more complete and meaningful view of ESG performance.

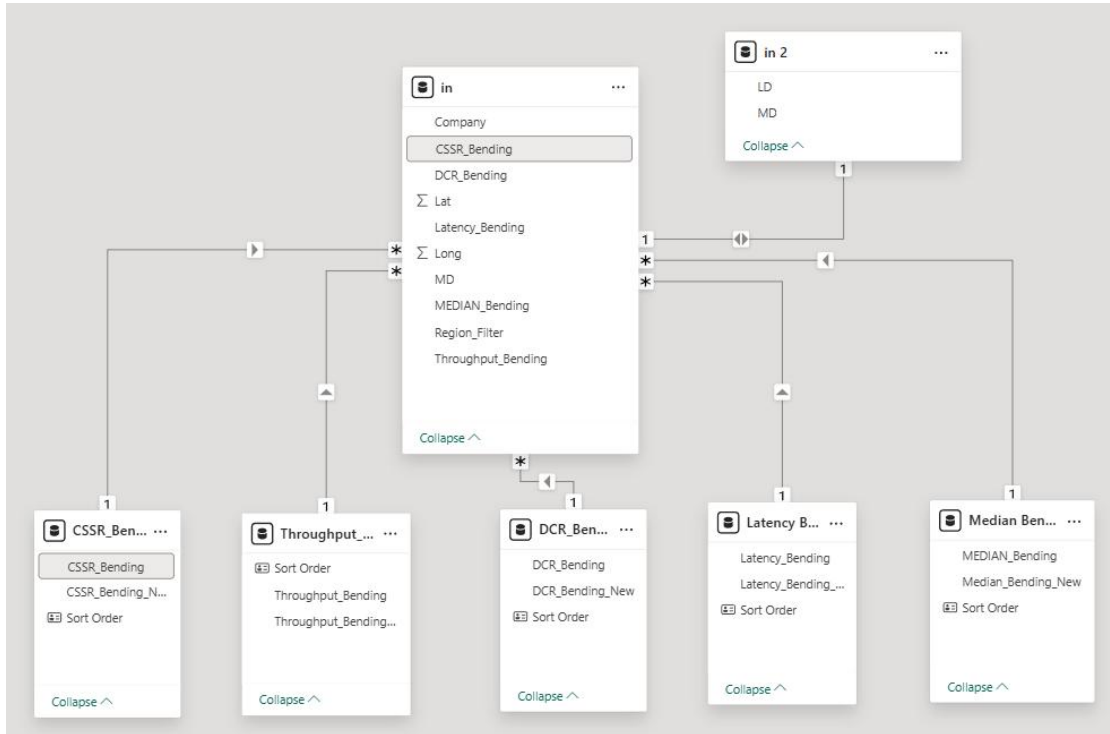


Figure 4 EMCE Data Modeling and Relationships

For the EMCE dashboard, the data model consists of two main tables, namely the “in” table and the “in 2” table. These tables are designed to support the analysis of mobile network performance and customer experience indicators.

Within the “in” table, several new calculated columns were created to categorize key performance metrics into meaningful groups. These include CSSR, Throughput, DCR, Latency, and Median bending columns. Each of these columns is used to classify raw numerical values into defined performance ranges (e.g. good or bad performance categories). This transformation simplifies data interpretation and enhances the effectiveness of visualizations.

Each of these newly created bending columns is then linked to corresponding reference tables (such as CSSR_Bending, Throughput_Bending, DCR_Bending, Latency_Bending, and Median_Bending) using one-to-many relationships. In this setup, the bending category table represents the “one” side, while the main “in” table represents the “many” side. This relationship

structure ensures consistent categorization and allows Power BI to apply proper filtering and grouping during analysis.

In addition, the “in 2” table is connected to the “in” table using the MD field through a one-to-one relationship. This indicates that each record in the “in” table corresponds directly to a single record in the “in 2” table. This type of relationship is used to extend the dataset with additional attributes without duplicating data, ensuring data integrity and consistency.

Overall, the data modeling approach for the EMCE dashboard ensures that performance metrics are well-structured, categorized, and efficiently connected. This enables accurate analysis of network performance and supports meaningful insights into customer experience.

5.5 Data Validation, Limitations, and Ethical Considerations

Ensuring the accuracy, reliability, and responsible handling of data is an essential component of the methodology used in this project. Since the dashboards developed in this project are intended to present analytical insights related to Environmental, Social, and Governance (ESG) performance as well as Enterprise Mobile Customer Experience (EMCE) metrics, it is important that the underlying data is processed correctly and that the visual outputs accurately represent the stored datasets. This section describes the procedures used to validate the data, outlines the limitations encountered during the project implementation, and discusses the ethical considerations related to the handling of organizational data.

A. Data Validation and Quality Assurance

To ensure the integrity and accuracy of the data used in the dashboards, several validation procedures were implemented throughout the development process. These validation procedures were applied during multiple stages of the data pipeline, including data preparation, data transformation, data modeling, and dashboard visualization.

During the data preparation stage, the simulated datasets created in Microsoft Excel were carefully reviewed to ensure that the data fields were structured correctly. Each dataset was designed with clearly defined columns representing specific performance indicators, such as sustainability metrics for ESG monitoring and network performance indicators for the EMCE dashboard. Before uploading the datasets into the storage environment, the structure of the Excel files was verified to ensure that all required columns were present and consistently named.

After the datasets were uploaded into BigQuery, additional validation checks were performed to confirm that the data had been successfully stored and could be accessed without errors. The stored datasets were then exported into CSV format and transferred to Microsoft Fabric, where further data processing was conducted. During this stage, the datasets underwent several data transformation and cleaning operations using Power Query to ensure that the data structure was appropriate for analytical modeling.

Several validation techniques were implemented to ensure data consistency and accuracy throughout the workflow:

- **Schema Validation:** Each dataset was reviewed to ensure that the structure of the tables matched the expected schema. Required columns such as performance indicators, timestamps, and location identifiers were verified to confirm that they were present and properly labeled.

- **Missing Data Detection:** The datasets were checked for missing or incomplete values that could potentially affect dashboard calculations. Any missing values identified during the transformation process were addressed through data cleaning procedures to ensure that the final dataset remained consistent.
- **Duplicate Data Identification:** Duplicate records were identified and removed during the data transformation stage to prevent incorrect aggregation results during dashboard calculations.
- **Data Type Verification:** Data fields were validated to ensure that they were assigned the correct data types, such as numerical values for performance indicators and text fields for categorical data. Proper data typing ensures that Power BI calculations and visualizations function correctly.
- **Relationship Validation:** Within the Power BI semantic model, relationships between multiple tables were verified to ensure that the connections between datasets were correctly established. This step ensures that aggregated calculations and visualizations are generated accurately across related datasets.
- **DAX Calculation Validation:** All DAX measures created within the Power BI model were tested to confirm that the calculated metrics accurately reflect the underlying datasets. The results produced by these calculations were cross-checked against the original dataset values to ensure accuracy.

Through these validation procedures, the project ensures that the developed dashboards provide reliable and accurate analytical insights. These validation steps help reduce the risk of displaying misleading information and ensure that the dashboards remain trustworthy tools for performance monitoring.

B. Limitations

Although the system architecture and dashboard design were developed to support effective data visualization and performance monitoring, several limitations were identified during the implementation of this project.

One of the primary limitations is the use of simulated datasets instead of real organizational data. Due to confidentiality and data protection policies within the organization, access to real ESG performance data and customer experience metrics was restricted. As a result, dummy datasets were created using Microsoft Excel to simulate realistic operational scenarios. While these datasets allow the dashboard functionality to be demonstrated effectively, they may not fully represent the complexity and scale of real operational data used within a production environment.

Another limitation relates to the data refresh mechanism used in the system architecture. The current data pipeline relies on scheduled data updates rather than real-time data streaming. In this implementation, the data stored in BigQuery is configured to refresh periodically, and the updated datasets are then integrated into Microsoft Fabric before being visualized in Power BI.

As a result, there may be a delay between when new data is generated and when it is reflected in the dashboard visualizations. This limitation means that the dashboards are primarily designed for periodic performance monitoring rather than real-time operational monitoring.

Additionally, the project focuses primarily on data visualization and analytical reporting rather than advanced predictive analytics or machine learning applications. While the dashboards provide valuable insights into performance trends and operational indicators, the system does not currently include predictive models that could forecast future performance trends or detect anomalies automatically.

Despite these limitations, the implemented system still demonstrates a functional data analytics pipeline that effectively transforms structured datasets into meaningful visual insights for performance monitoring.

C. Ethical and Data Privacy Considerations

Ethical considerations and data privacy protection are important aspects of any data analytics project, particularly when dealing with corporate data and customer-related information. In this project, several measures were taken to ensure that the development process adheres to responsible data handling practices.

The most important ethical consideration in this project is the protection of confidential organizational data. Since the dashboards are based on operational performance metrics related to telecommunications services, the actual corporate datasets contain sensitive information that cannot be disclosed outside the organization. To address this concern, the project uses simulated datasets created specifically for development and demonstration purposes. These dummy datasets replicate the structure of real operational data while ensuring that no sensitive information is exposed.

Using simulated datasets ensures that customer privacy and organizational confidentiality are maintained throughout the development process. This approach aligns with corporate data governance policies that restrict the external sharing of internal operational data.

Another ethical consideration relates to the responsible presentation of analytical results. When designing dashboards and visualizations, it is important to ensure that the information presented is clear, accurate, and not misleading. During the dashboard development process, careful

attention was given to selecting appropriate visualization types and ensuring that the displayed metrics accurately represent the underlying datasets.

Furthermore, access to the dashboards should be controlled to ensure that only authorized personnel can view sensitive performance metrics. Although the project focuses primarily on dashboard development rather than access management systems, proper governance mechanisms should be implemented in real-world deployment environments to ensure that data access is restricted to relevant stakeholders.

Overall, the use of simulated datasets, careful data validation procedures, and responsible visualization practices ensures that the project adheres to ethical data handling standards while still demonstrating the capabilities of modern business intelligence systems.

6.0 RESULTS AND DISCUSSION

The final outcome of this project consists of two interactive dashboards developed using Microsoft Power BI: the ESG Dashboard and the Enterprise Mobile Customer Experience (EMCE) Dashboard. The ESG dashboard enables centralized monitoring of sustainability-related indicators, allowing organizations to track environmental and governance metrics while improving transparency in sustainability reporting. Such monitoring tools support corporate accountability and performance evaluation (Buallay & Al Marri, 2022).

The EMCE dashboard focuses on monitoring mobile network performance and customer experience. Key indicators such as call setup success rate, drop call rate, streaming latency, and streaming throughput are visualized through charts and maps. These visualizations help management teams identify network performance issues and improve service quality to maintain customer satisfaction (Mogire et al., 2024). Overall, the dashboards demonstrate how business intelligence tools can transform raw data into meaningful insights that support operational decision-making..

All screenshots and data shown are based on a mock interface and dummy data, provided for illustration purposes only.

6.1 ESG Dashboard Result (SAP)

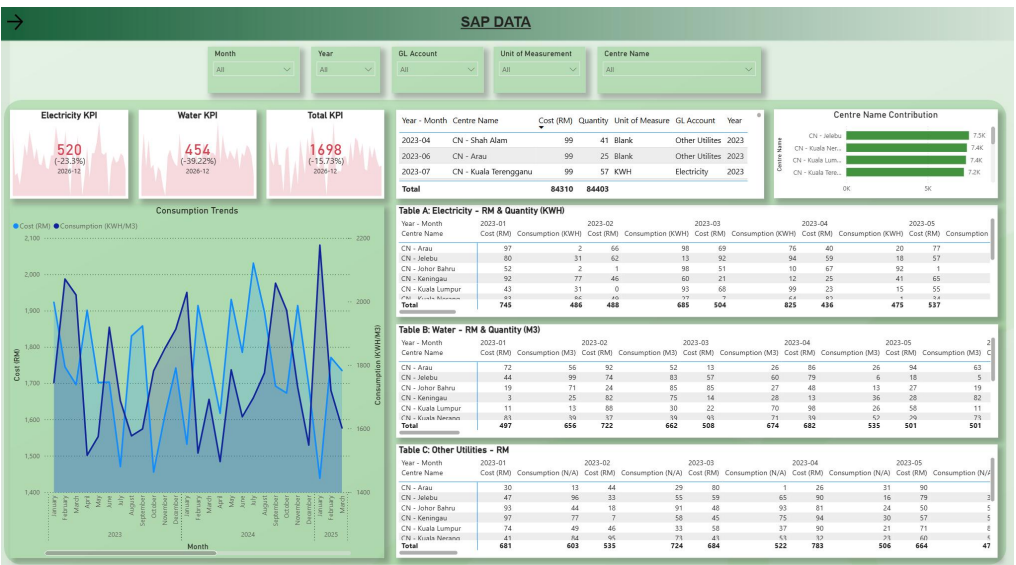


Figure 6.1 ESG (SAP Dashboard, Dummy data for illustration purpose)

Figure 6.1 presents the main interface of the ESG dashboard developed using Power BI. The dashboard is designed to provide a centralized platform for monitoring sustainability-related indicators such as electricity consumption, water usage, and other utility metrics across different operational centres. The dashboard integrates several visualizations including bar charts, line charts, and summary indicators that allow users to quickly interpret the environmental performance of the organization.

Several slicers are available on the dashboard, including **Month, Year, GL Account, Unit of Measurement, and Centre Name**. These slicers allow users to dynamically filter the dataset based on specific parameters. For instance, the Month slicer allows users to view consumption data for any selected month from January to December, while the Year slicer allows comparison across different years ranging from 2023 to 2026. This flexibility enables users to analyze seasonal trends and identify patterns in energy or utility usage over time.

In addition, the GL Account slicer enables users to filter data based on specific utility categories such as electricity, water, or other utilities. This helps management understand which operational areas contribute the most to resource consumption. The Centre Name slicer allows users to analyze consumption at different operational locations such as Kuala Lumpur, Johor Bahru, Keningau, and Arau. By comparing different centres, decision-makers can identify locations with higher resource consumption and implement strategies to improve efficiency and sustainability performance.

Overall, this dashboard improves transparency and enables organizations to monitor environmental performance more effectively, supporting sustainability initiatives and data-driven decision-making.

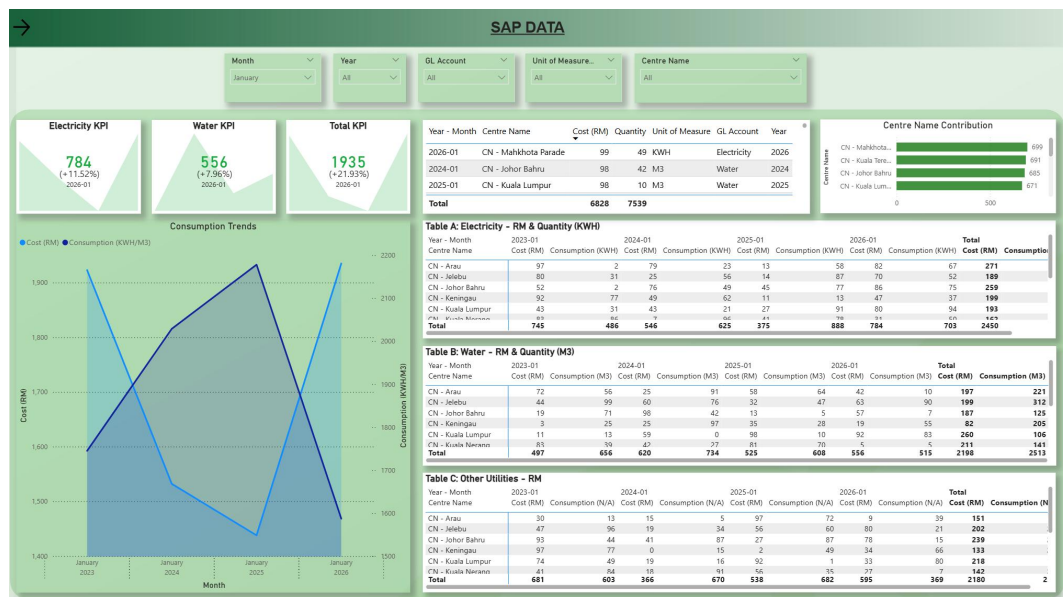


Figure 6.2 ESG (SAP Dashboard filter by Month, Dummy data for illustration purpose)

Figure 6.2 illustrates the dashboard when the Month slicer is applied. The Month slicer allows users to filter the dataset from January to December, arranged in chronological order. When a specific month is selected, all charts and visual elements within the dashboard automatically update to display data corresponding to that selected month.

This functionality allows users to analyze monthly patterns in utility consumption across different centres. For example, users can observe whether electricity or water usage increases during certain months due to higher operational activities. By comparing different months, organizations can identify seasonal trends and better understand how resource consumption changes throughout the year.

Such analysis can assist management in planning operational activities more efficiently and implementing strategies to reduce resource consumption during peak periods.

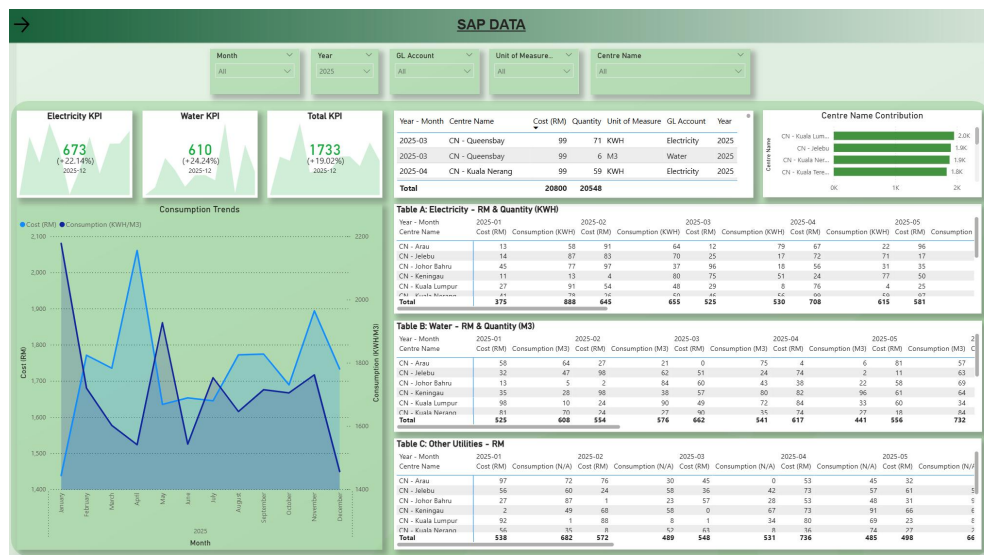


Figure 6.3 ESG (SAP Dashboard filter by Year, Dummy data for illustration purpose)

Figure 6.3 presents the dashboard when the Year slicer is applied. The Year slicer allows users to select data from 2023 until 2026, arranged from the most recent year to the oldest. Once a specific year is selected, the dashboard dynamically updates to display all visualizations based on the chosen year.

This feature enables users to perform year-to-year comparisons and evaluate long-term trends in utility consumption. By analyzing the changes in electricity and water usage across different years, organizations can assess whether their sustainability initiatives have been effective. For example, a decrease in energy consumption over time may indicate improved operational efficiency or successful implementation of energy-saving practices.

Therefore, the Year filter helps organizations monitor performance over time and supports strategic decision-making related to sustainability and resource management.



Figure 6.4 ESG (SAP Dashboard filter by GL Account, Dummy data for illustration purpose)

Figure 6.4 shows the ESG dashboard when the GL Account slicer is applied. The GL Account filter allows users to view data based on specific utility categories such as Electricity, Water, and Other Utilities. By selecting a particular GL Account, the dashboard displays only the data related to that specific utility category.

This feature allows users to focus on individual types of resource consumption and analyze them separately. For instance, when the Electricity account is selected, the dashboard highlights electricity usage patterns across different centres and time periods. This helps management identify areas where electricity consumption is particularly high and investigate potential causes.

By isolating different utility categories, organizations can better understand their resource usage and develop targeted strategies to reduce consumption and improve operational efficiency.

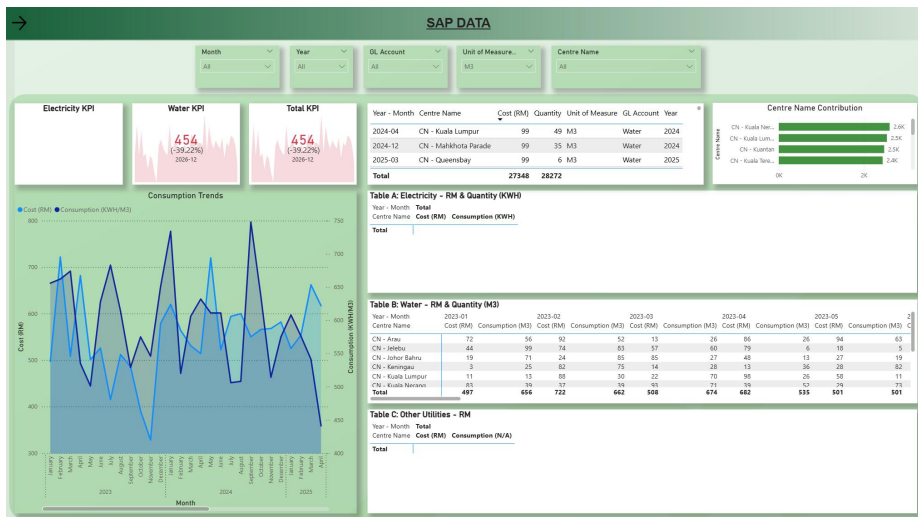


Figure 6.5 ESG (SAP Dashboard filter by Unit of Measurement, Dummy data for illustration purpose)

Figure 6.5 illustrates the dashboard when the Unit of Measurement slicer is applied. The available units include KWH for electricity consumption and M3 for water consumption, along with a blank option for records without a specified measurement unit.

This filter allows users to analyze the data according to the measurement unit used for different utilities. For example, selecting KWH will display electricity consumption data measured in kilowatt-hours, while selecting M3 will show water consumption measured in cubic meters. This helps ensure that users interpret the data accurately based on the appropriate unit of measurement.

By organizing data according to measurement units, the dashboard improves the clarity of analysis and helps users focus on specific types of resource consumption.



Figure 6.6 ESG (SAP Dashboard filter by Centre Name, Dummy data for illustration purpose)

Figure 6.6 presents the dashboard when the Centre Name slicer is applied. The available centres include locations such as Mahkhota Parade, Keningau, Johor Bahru, and Kuala Lumpur. When a particular centre is selected, the dashboard updates to display utility consumption data associated only with that specific location.

This functionality allows organizations to analyze the performance of individual operational centres. By comparing resource usage across different locations, management can identify centres with higher consumption levels and investigate possible reasons such as operational demand, infrastructure conditions, or equipment efficiency.

Through this analysis, organizations can implement location-specific improvements to optimize resource usage and reduce unnecessary operational costs.

6.2 ESG Dashboard Result (NoN - SAP)

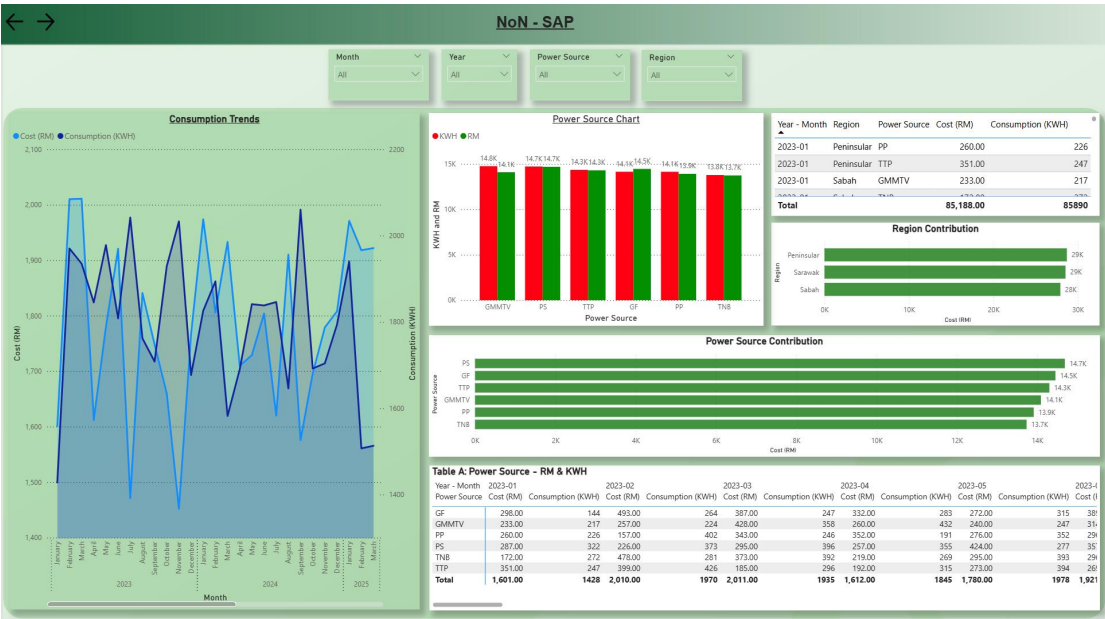


Figure 6.7 ESG (NoN - SAP Dashboard, Dummy data for illustration purpose)

Figure 6.7 presents the overall view of the ESG dashboard developed using Non-SAP data. This dashboard focuses on visualizing electricity consumption and energy-related metrics across different regions and power sources. The interface includes several interactive slicers such as Month, Year, Power Source, and Region, which allow users to dynamically explore the dataset.

The overall dashboard provides a summarized visualization of electricity usage, enabling users to quickly understand how energy consumption is distributed across different categories. The combination of charts and graphical elements allows users to identify general patterns, such as which regions or power sources contribute the most to total consumption. This overview serves as an entry point for users before applying more specific filters to perform deeper analysis.

By providing a centralized and interactive view, the dashboard supports better monitoring of energy usage and helps organizations gain a clearer understanding of their overall consumption behavior.

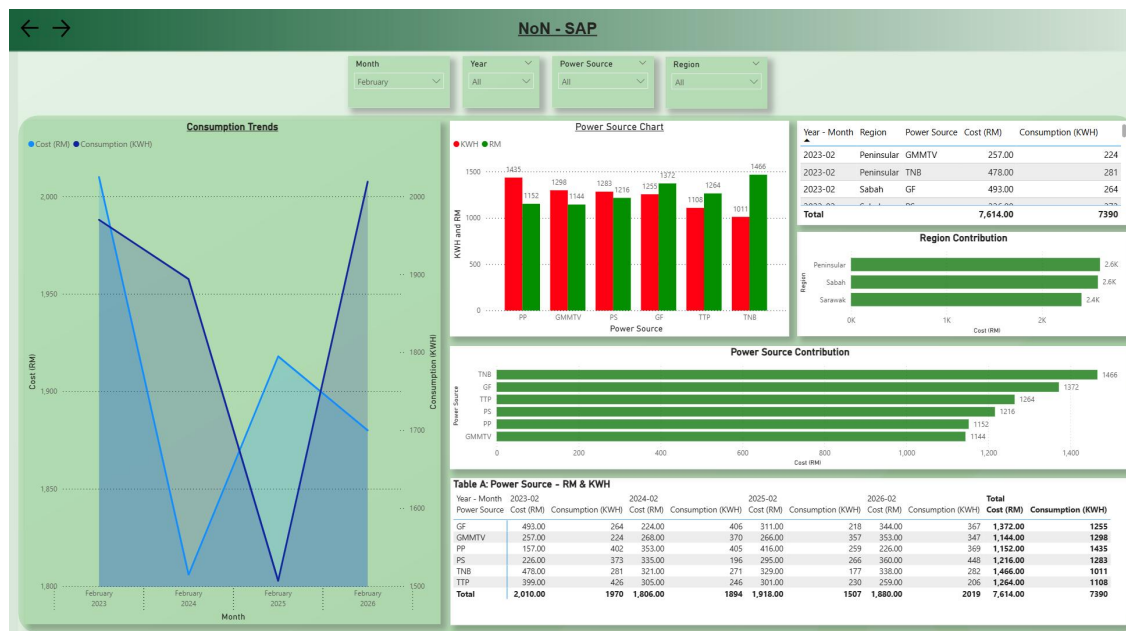


Figure 6.8 ESG (NoN - SAP Dashboard filter by Month, Dummy data for illustration purpose)

Figure 6.8 illustrates the dashboard when the Month slicer is applied. Users can select any month from January to December, arranged in chronological order. Once a specific month is selected, all visualizations within the dashboard automatically update to reflect the electricity consumption data for that particular month.

This functionality enables users to analyze monthly fluctuations in energy usage and identify patterns that may be influenced by operational activities or seasonal factors. For example, certain months may show higher electricity consumption due to increased business operations or environmental conditions.

By comparing data across different months, organizations can detect peak usage periods and plan accordingly to optimize energy consumption. This helps in improving operational efficiency and reducing unnecessary energy costs.

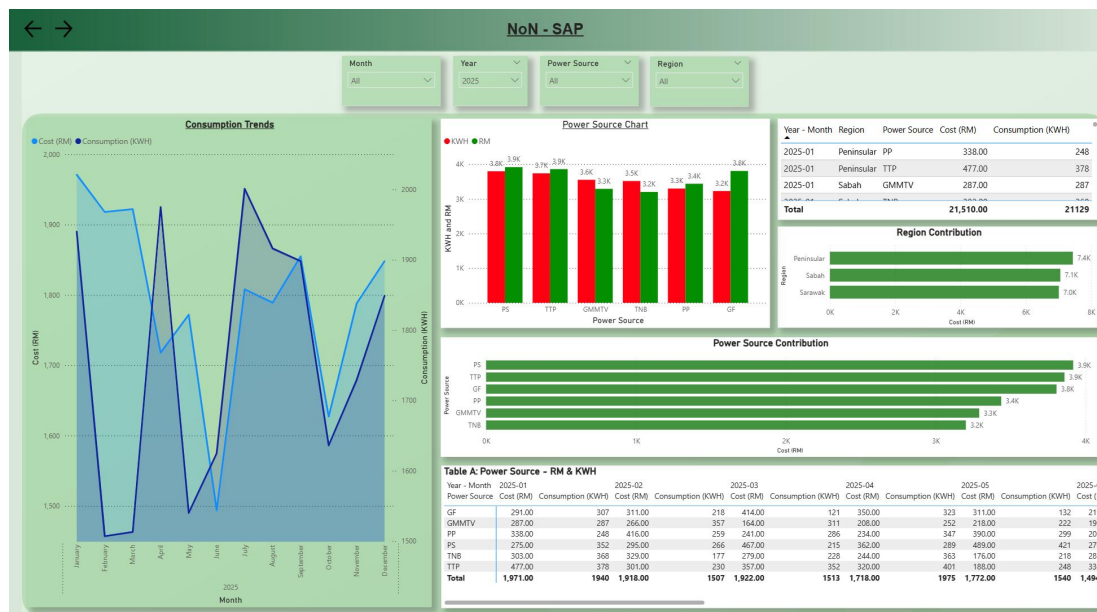


Figure 6.9 ESG (NoN - SAP Dashboard filter by Year, Dummy data for illustration purpose)

Figure 6.9 shows the dashboard when the Year slicer is applied. The available years range from 2023 to 2026, arranged from the most recent to the oldest. When a specific year is selected, all charts and visual elements update to display the corresponding data.

This feature allows users to perform year-to-year comparisons and observe long-term trends in electricity consumption. By analyzing how energy usage changes over time, organizations can evaluate the effectiveness of energy management strategies and sustainability initiatives.

For instance, a decrease in electricity consumption in later years may indicate improved efficiency or successful implementation of cost-saving measures. Therefore, the Year filter supports strategic decision-making by providing insights into long-term performance trends.

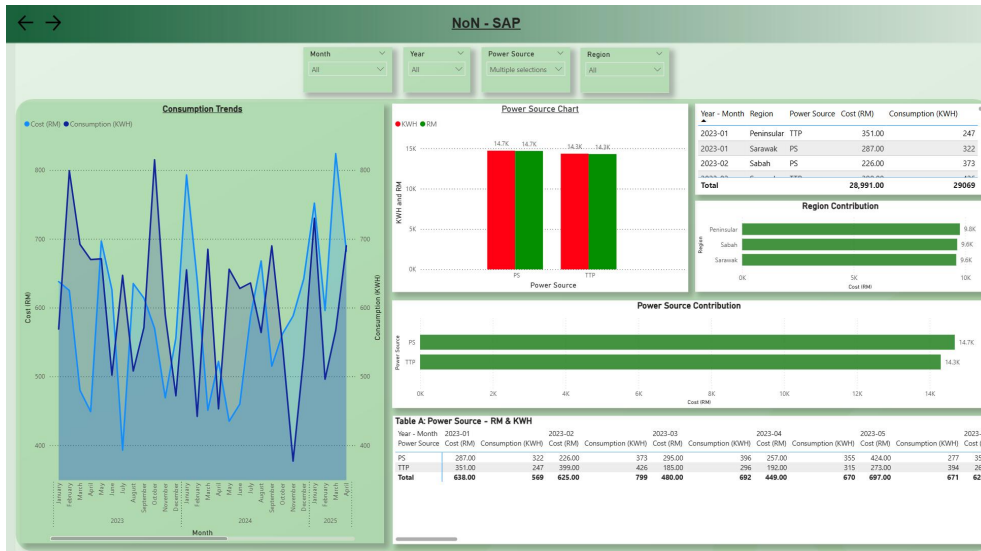


Figure 6.10 ESG (NoN - SAP Dashboard filter by Power Source, Dummy data for illustration purpose)

Figure 6.10 presents the dashboard when the Power Source slicer is used. The available power sources include GF, GMMTV, PP, PS, TNB, and TTP. Selecting a specific power source will update the dashboard to display electricity consumption data associated with that source only.

This filtering capability allows users to analyze how electricity is distributed among different energy providers. It helps organizations identify which power sources contribute the most to overall consumption and assess their dependency on specific providers.

Such insights are important for energy planning and management, as organizations can evaluate whether certain power sources are more efficient or cost-effective. This can support better decision-making in selecting or optimizing energy supply sources.

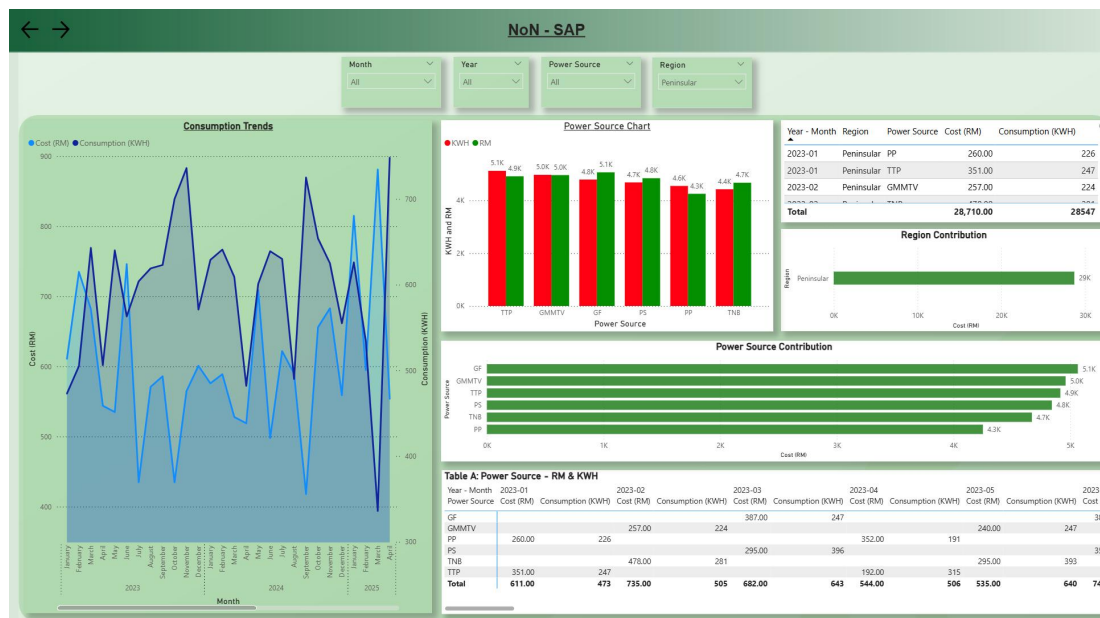


Figure 6.11 ESG (NoN - SAP Dashboard filter by Region, Dummy data for illustration purpose)

Figure 6.11 shows the dashboard when the Region slicer is applied. Users can filter electricity consumption data based on geographical regions such as Peninsular Malaysia, Sabah, and Sarawak. When a region is selected, the dashboard updates to display data corresponding only to that specific area.

This functionality enables organizations to compare electricity usage across different regions and identify geographical patterns in energy consumption. For example, certain regions may show higher electricity usage due to larger operational activities or infrastructure demands.

By analyzing regional differences, management can detect inefficiencies and implement targeted improvements in specific areas. This helps optimize resource usage and ensures that energy management strategies are applied effectively across all regions.

6.3 EMCE Dashboard Result

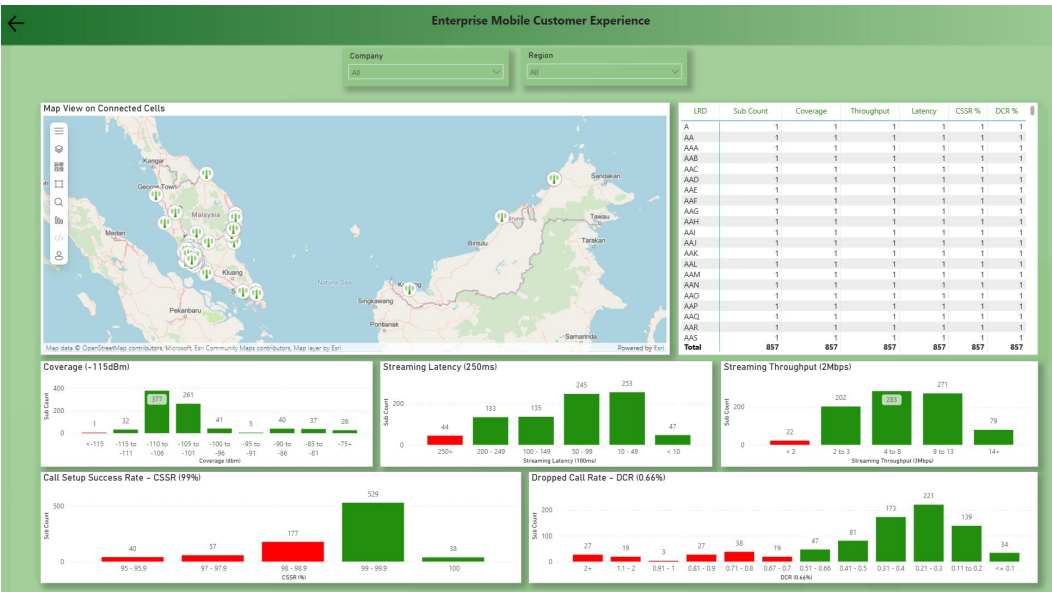
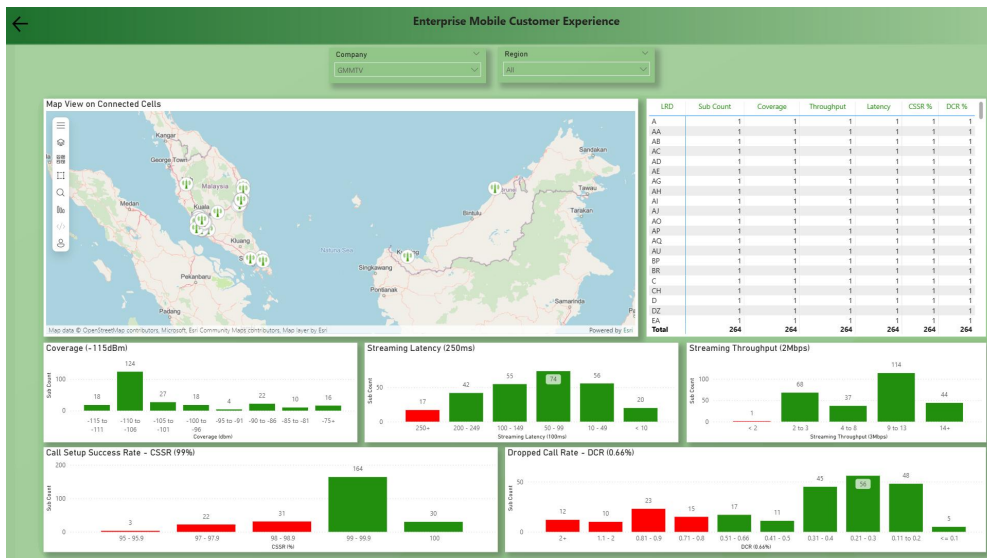


Figure 6.12 EMCE (Dummy data for illustration purpose)

Figure 6.12 presents the overall view of the Enterprise Mobile Customer Experience (EMCE) dashboard developed using Microsoft Power BI. This dashboard is designed to monitor mobile network performance and evaluate customer experience across different companies and regions. The interface includes slicers such as **Company (DMD, GMMTV, KBS, and SBS)** and **Region (HYBE, JYP and SM)**, allowing users to dynamically filter and explore the data.

The dashboard visualizes key performance indicators (KPIs) including **Coverage**, **Latency**, **Throughput**, **Call Setup Success Rate (CSSR)**, and **Drop Call Rate (DCR)**. These indicators are essential in assessing the quality and reliability of mobile network services. For example, coverage values greater than -115 indicate poor signal strength, latency above 250ms reflects delays in data transmission, and throughput below 2mbps indicates weak network performance. In contrast, CSSR values greater than or equal to 99 and DCR values less than or equal to 0.66 represent good network performance.

Overall view, users can quickly identify the general performance of the network across all companies and regions. This helps management gain a high-level understanding of service quality and detect whether there are widespread performance issues. The dashboard supports faster decision-making by transforming raw data into meaningful visual insights that highlight strengths and weaknesses in network performance.



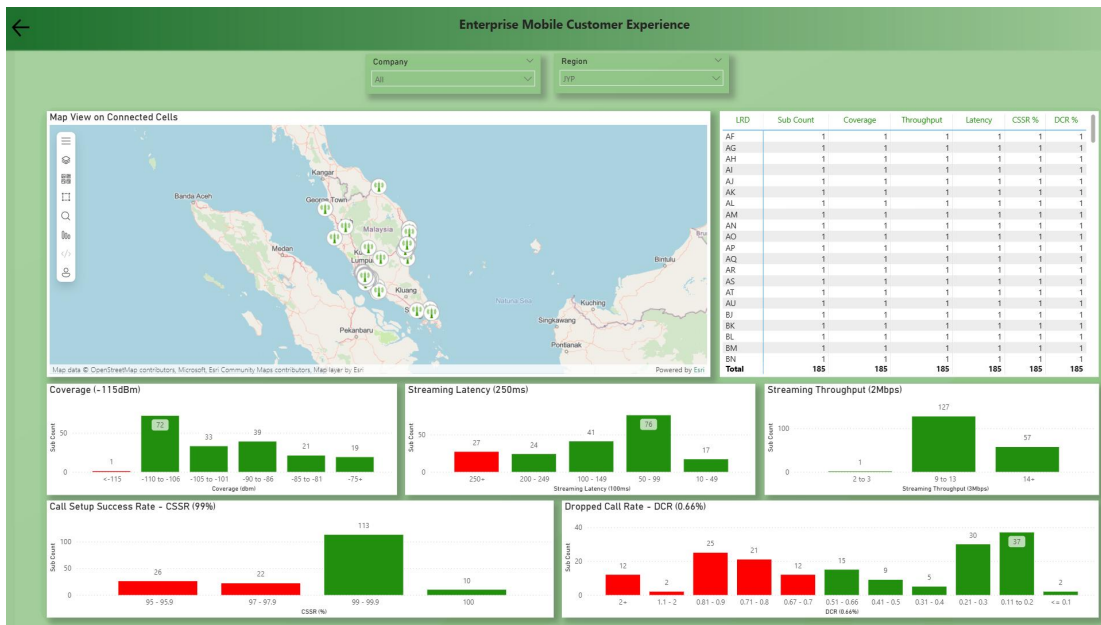


Figure 6.14 EMCE (Filter by Region, Dummy data for illustration purpose)

Figure 6.14 illustrates the dashboard when the Region slicer is applied. The regions include **SM**, **HYBE**, and **JYP**. When a region is selected, the dashboard updates to reflect network performance data specific to that geographical area.

This feature enables users to analyze and compare network performance across different regions. It helps identify whether certain areas are experiencing weaker coverage, higher latency, or lower throughput. For example, regions with more rural or remote locations may show poorer performance compared to urban areas.

By examining regional differences, management can identify areas that require improvement and allocate resources more effectively. This supports better network optimization and ensures that service quality is maintained across all regions. Ultimately, the ability to monitor performance at the regional level helps improve overall customer experience and ensures consistent service delivery.

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